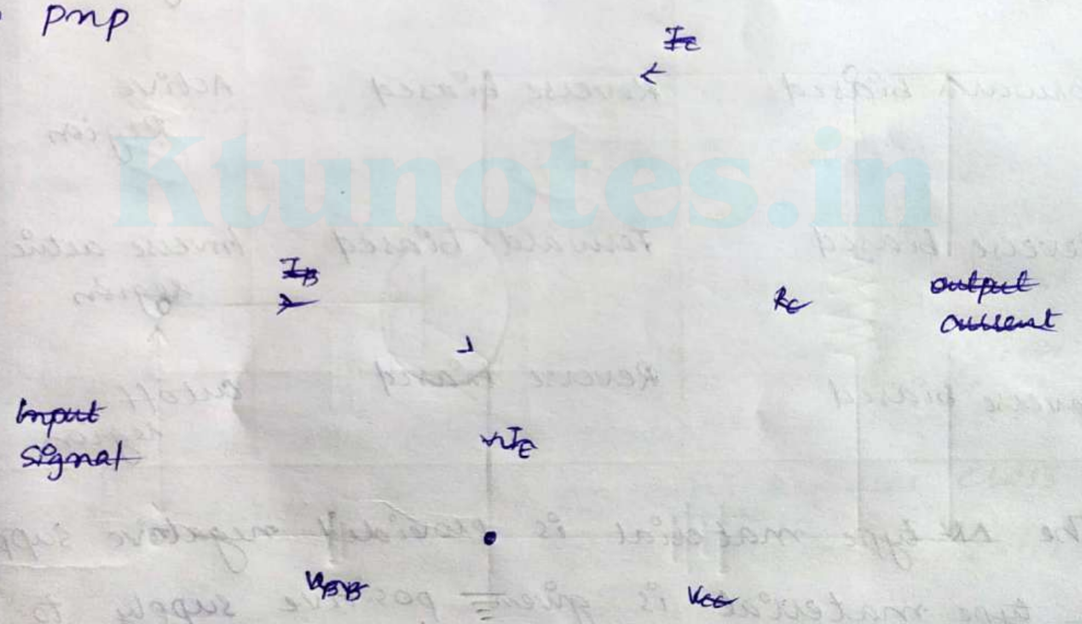
MODULE - 1

BIPOLAR JUNCTION TRANSISTORS (BJTs)

* A bipolar junction transistor (BJT) is a type of transistor that uses both electrons and holes as charge carriers.
* Mainly used for amplification
* Two types :
    - npn
    - pnp



Three terminals

1. Emitter : Heavily doped
2. Base : Lightly doped
3. collector : Moderately doped

Two junctions

1. Emitter - base junction
2. Collector - base junction

Operation Regions of a transistor

EMITTER JUNCTION	COLLECTOR JUNCTION	REGION OF OPERATION	APPLICATION
Forward biased	Forward biased	Saturation region	Closed switch
Forward biased	Reverse biased	Active Region	Amplifier
Reverse biased	Forward biased	Inverse active region	
Reverse biased	Reverse biased	cutoff region	Open switch

* The N-type material is provided negative supply and p-type material is given positive supply to make the circuit Forward bias
* The N-type material is provided positive supply and p-type material is given negative supply to make the circuit Reverse bias.

Operation of NPN Transistor

- Emitter - base junction is forward biased and collector - base junction is reverse biased.

* The voltage V provides a negative potential at the emitter which repels the electron in the N- type material and these electrons cross the emitter -base junction & reach the base region. There is very low percent of electrons recombine with free holes of P-region. This provides very low current which constitute the base current I_B. The remaining holes cross the collector - base junction, constitute the collector current I_C.

* Emitter current = Base current + collector current

$$I_E = I_B + I_C$$

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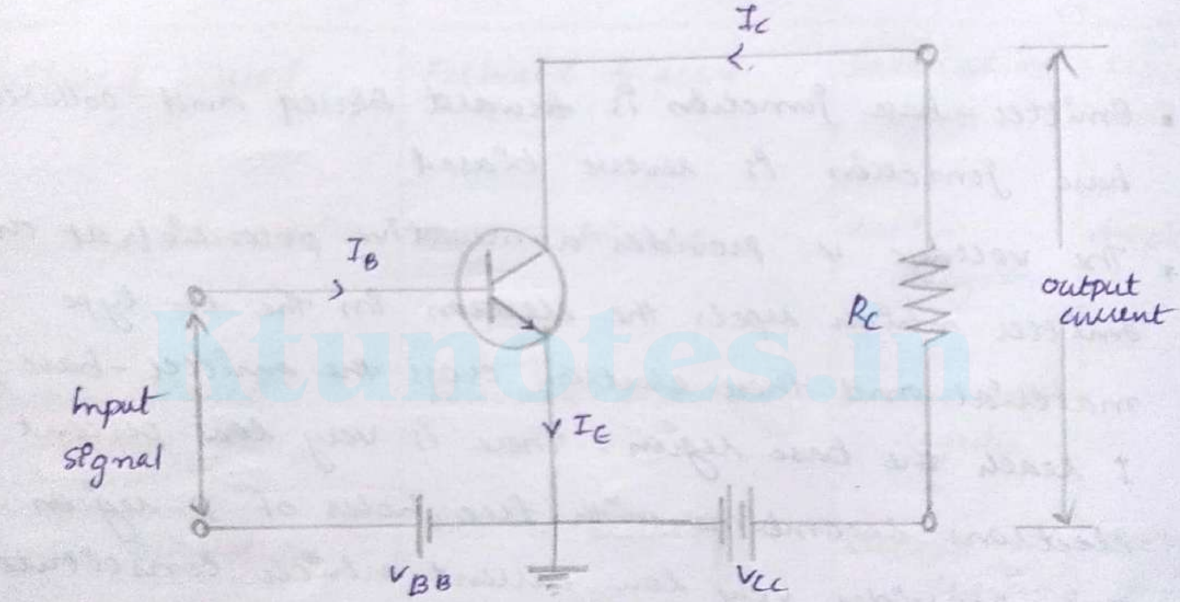
BJT CONFIGURATIONS

Three configurations

* Common Emitter Configuration
* Common Base configuration
* Common collector configuration

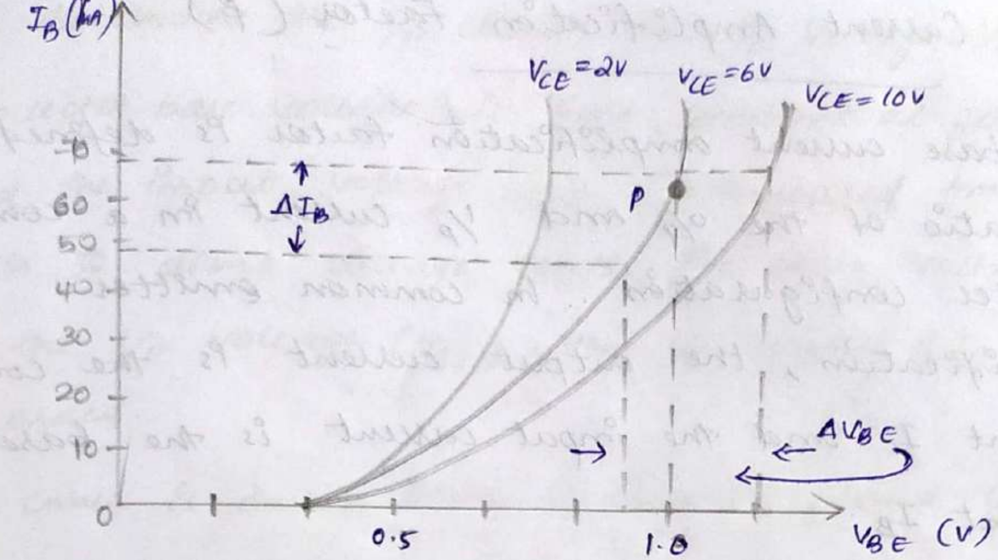
NPN transistor

COMMON EMITTER CONFIGURATION



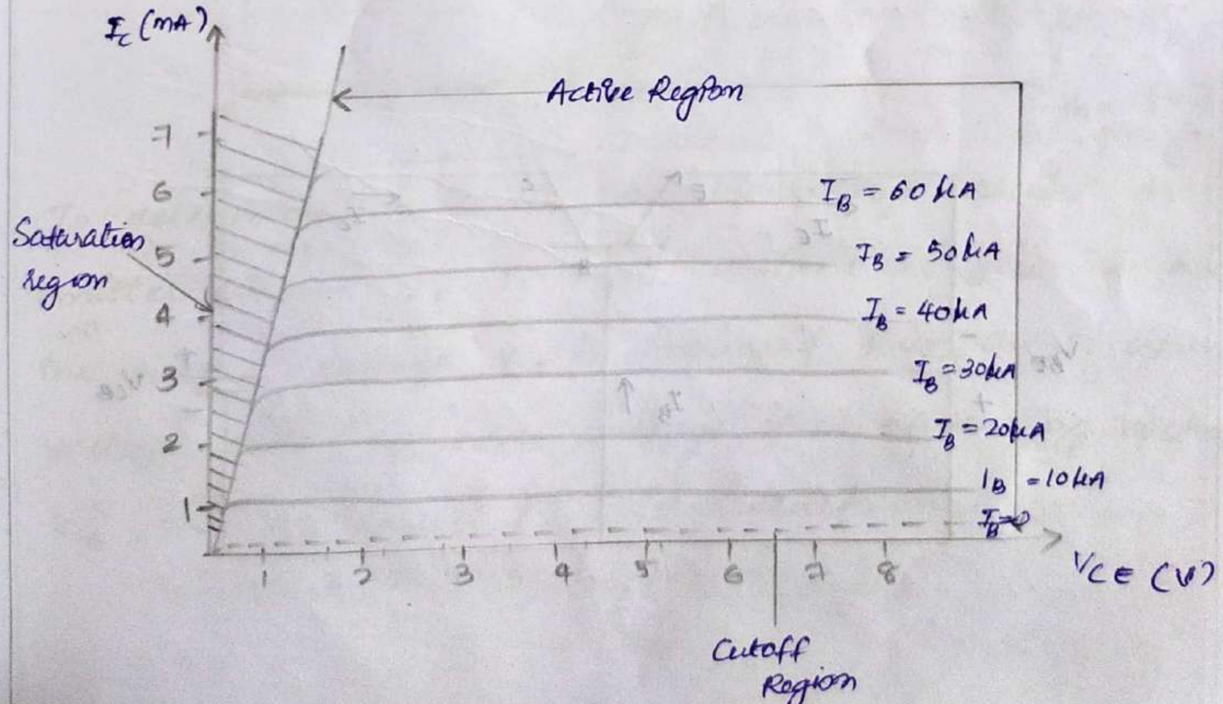
Input Characteristic Curve

The curve plotted between base current I_B and the base-emitter voltage V_{EB} is called input characteristics curve. For drawing the I/p chara the reading of base currents is taken through the ammeter on emitter voltage V_{BE} at constant collector-emitter current.



Output Characteristic

In CE configuration, the curve draws between collector current I_C and collector-emitter voltage V_{CE} at a constant base current I_B is called o/p characteristic. The characteristic curve for the typical NPN transistor in CE configuration



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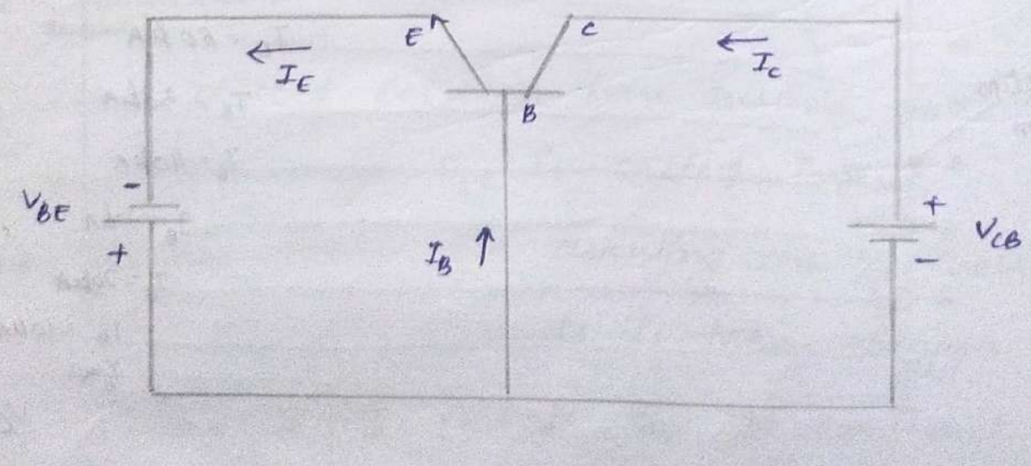
Base Current Amplification factor (β)

The base current amplification factor is defined as the ratio of the o/p and i/p current in a common emitter configuration. In common emitter amplification, the output current is the collector current I_C and the input current is the base current I_B.

In other words, the ratio of change in collector current w.r.t base current is known as the base amplification factor. It is represented by β.

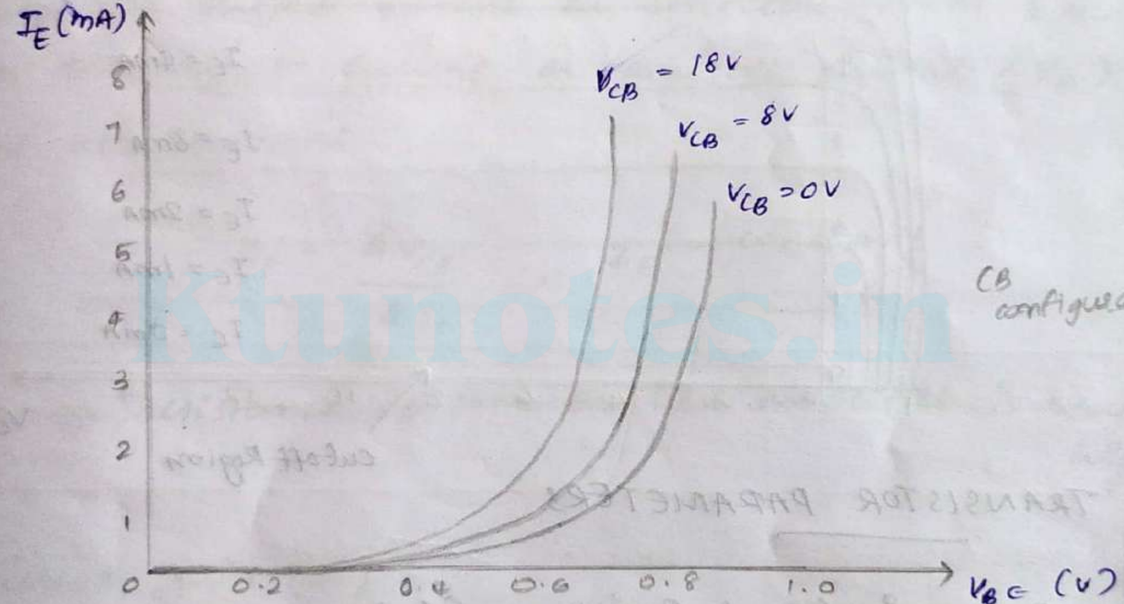
$$\beta = \frac{\Delta I_C}{\Delta I_B}$$

COMMON BASE CONFIGURATION



To determine the i/p chara, the o/p voltage V_{CB} (collector base voltage) is kept constant at zero volts and the input voltage V_{BE} is increased from zero volts to dfrnt voltage levels. For each voltage level of the i/p voltage (V_{BE}), the i/p current (I_E) is recorded.

A curve is drawn btwn i/p current I_E and i/p voltage V_{BE} at constant o/p voltage V_{CB} (0 volts).



CB configuration

To determine o/p chara, the i/p ~~voltage~~ current or emitter current I_E is kept constant at zero mA and the output voltage V_{CB} is increased from 0V to dfrnt voltage levels. For each voltage level of the o/p voltage V_{CB}, the o/p current (I_C) is recorded.

A curve is drawn btwn o/p current I_C and output voltage V_{CB} at constant i/p current I_E (0mA) when the emitter current or i/p current I_E is equal to 0mA, the transistor operates in the cut-off region.

TRANSISTOR PARAMETERS

* Dynamic Input Resistance (r_i) :

It is defined as the ratio of change in input voltage or emitter voltage (V_{BE}) to the corresponding change in input current or emitter current (I_E), with the o/p voltage or collector voltage (V_{CB}) kept at constant.

$$r_i = \frac{\Delta V_{BE}}{\Delta I_E}, \quad V_{CB} = \text{constant}$$

The i/p resistance of common base amplifier is very low.

* Dynamic Output Resistance (r_o):

It is defined as the ratio of change in output voltage or collector voltage (V_{CB}) to the corresponding change in output current or collector current (I_C), with the input current or emitter current (I_E) kept at constant.

$$r_o = \frac{\Delta V_{CB}}{\Delta I_C}, \quad I_E = \text{constant}$$

The o/p resistance of common base amplifier is very high

* Current Gain (α):

The current gain of a transistor in CB configuration is defined as the ratio of output current or collector current (I_C) to the input current or emitter current (I_E)

$$\alpha = \frac{I_C}{I_E}$$

The current gain of a transistor in CB configuration is less than unity. The typical current gain of common base amplifier is 0.98

Relationship Between amplification factors α and β

Using $\beta = \frac{I_C}{I_B}$, $\alpha = \frac{I_C}{I_E}$

and $I_E = I_C + I_B$

$$\frac{I_C}{\alpha} = I_C + \frac{I_C}{\beta} \rightarrow \frac{1}{\alpha} = 1$$

$$\beta = \alpha\beta + \alpha = (\beta + 1)\alpha$$

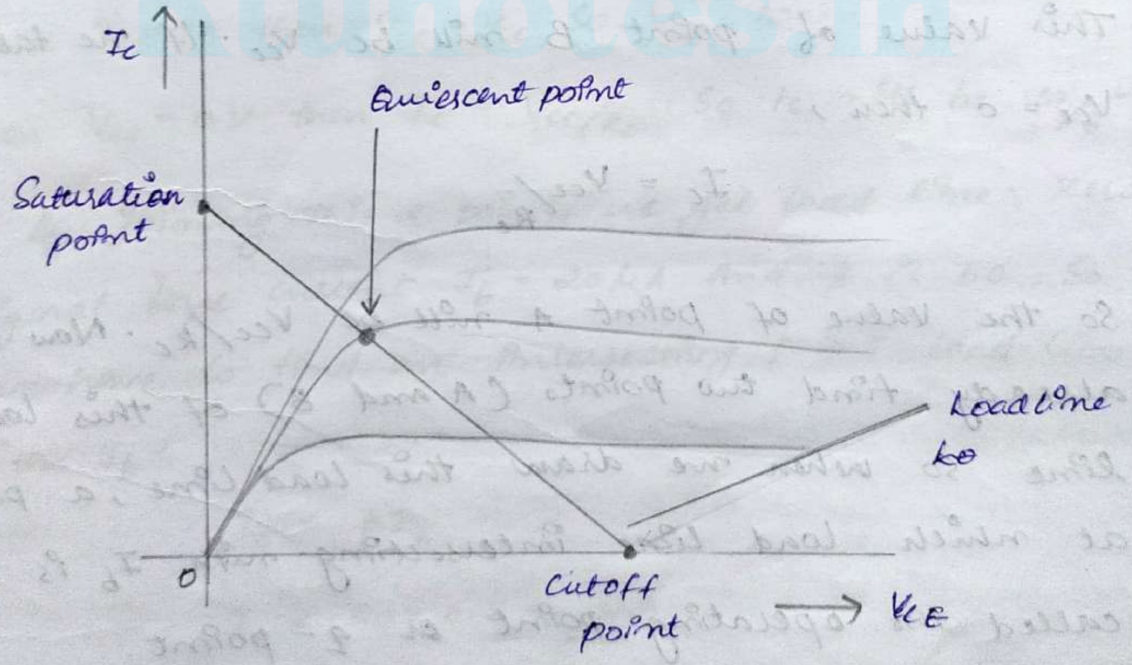
$$\boxed{\alpha = \frac{\beta}{\beta + 1}} \qquad \boxed{\beta = \frac{\alpha}{\alpha - 1}}$$

Characteristic	Common Base	Common Emitter	Common collector
Input impedance	Low	Medium	High
Output impedance	Very high	High	Low
Phase shift	0°	180°	0°
Voltage gain	High	Medium	Low
Current gain	Low	Medium	High
Power gain	Low	Very high	Medium

Load lines and Q point of a Transistor

When a line is drawn joining the saturation and cut off points, such a line can be called as Load line. This line, when drawn over the output characteristic curve, makes contact at a point called as operating point.

This operating point is also called as quiescent point or simply Q-point. There can be many such intersecting points, but the Q-point is selected in such a way that irrespective of AC signal swing, the transistor remains in the active region.



For finding operating point first we have to find load line points (A and B) on I_C and V_{CE}.
For general transistor biasing circuit, output circuit equation is

$$V_{CE} = V_{CC} - I_C R_C$$

The o/p chara of this transistor show graph between I_C and V_{CE}. For finding intersecting points of load line with x-axis and y-axis we take one by one $I_C = 0$ and then $V_{CE} = 0$

1st we take $I_C = 0$ sothat

$$V_{CE} = V_{CC}$$

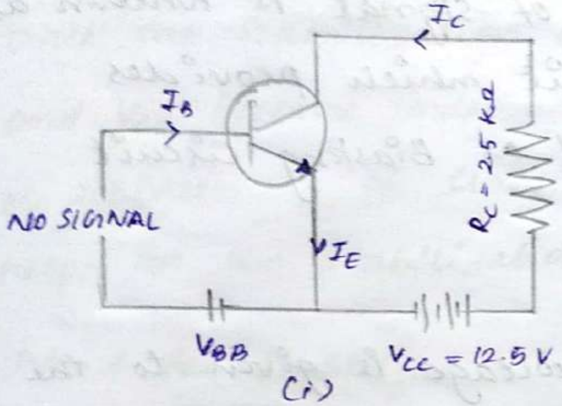
This value of point B will be V_{CC}. If we take $V_{CE} = 0$ then,

$$I_C = V_{CC}/R_C$$

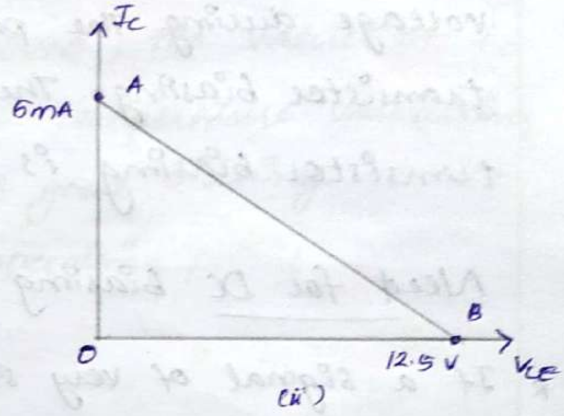
So the value of point A will be V_{CC}/R_C. Now we already find two points (A and B) of this load line. So when we draw this load line, a point at which load line intersecting with I_b is called the operating point or Q-point.

A) As the circuit shown below

If $V_{cc} = 12V$ and $R_C = 6K\Omega$, draw the d.c load line. What will be the Q-point if zero signal base current is $20\mu A$ and $\beta = 50$



(i)



(ii)

Here collector - emitter voltage V_{ce} is given by

$$V_{ce} = V_{cc} - I_C R_C$$

When $I_C = 0A$ then $V_{ce} = V_{cc}$ so $V_{ce} = 12V$

when $V_{ce} = 0V$ then $I_C = V_{cc}/R_C$. So I_C will be $12/6 =$

By joining this 2 points we get load line. Zero signal base current $I_b = 20\mu A$ and β is 50. So we have to find an intersecting point load line with I_b.

$$I_C = \beta I_b$$

$$I_C = 50 \times 0.2 = \underline{1mA}$$

Zero signal collector - emitter voltage is

$$V_{ce} = V_{cc} - I_C R_C = 12 - (1mA \times 6K\Omega) = \underline{\underline{6V}}$$

TRANSISTOR BIASING

The proper flow of zero signal collector current and the maintenance of proper collector emitter voltage during the passage of signal is known as transistor biasing. The circuit which provides transistor biasing is called as Biasing circuit.

Need for DC biasing

* If a signal of very small voltage is given to the i/p of BJT, it cannot be amplified. Because, for a BJT, ~~it cannot be~~ to amplify a signal, two conditions have to be met.
* The input voltage should exceed cut in voltage for the transistor to be ON.
* The BJT should be in the active region, to be operated as an amplifier.
* If appropriate DC voltages and currents are given through BJT by external sources, so that BJT operates in active region and superimpose the AC signals to be amplified, then this problem can be avoided. The given DC voltage and currents are so chosen that the transistor remains in active region for entire input AC cycle. Hence DC biasing is needed.

STABILIZATION

* The process of making the operating point independent of temperature changes or variations in transistor parameters is known as stabilization.
* Once the stabilization is acheived, the values of I_C and V_{CE} become independent of temperature variations or replacement of transistor. A good biasing circuit helps in the stabilization of operating point.

Need for Stabilization

Stabilization of the operating point has to be acheived due to the following reasons:

* Temperature dependence of I_C
* Individual Variations
* Thermal runway

Temperature Dependence of I_C

As the expression for collector current I_C is

$$I_C = \beta I_B + I_{CEO}$$

$$= \beta I_B + (\beta+1) I_{CBO}$$

The collector leakage current I_{CBO} is greatly influenced by temperature variations. To come out of this, the biasing conditions are set so that zero signal collector current $I_C = 1mA$. Therefore, the operating point needs to be stabilized. i.e; it is necessary to keep I_C constant.

Individual Variations

As the value of β and the value of V_{BE} are not same for every transistor, whenever a transistor is replaced, the operating point tends to change. Hence it is necessary to ~~keep I_C constant~~ stabilize the operating point.

Thermal Runaway

As the expression for collector current I_C is

$$I_C = \beta I_B + I_{CEO}$$

$$= \beta I_B + (\beta+1) I_{CBO}$$

The flow of collector current and also the collector leakage current causes heat dissipation. If the operating point is (not) stabilized, there occurs a cumulative effect which increases this heat dissipation.

The self-destruction of such an unstabilized transistor is known as Thermal runway. In order to avoid thermal runway and the destruction of transistor, it is necessary to stabilize the operating point, i.e., to keep I_C constant.

STABILITY FACTOR

* It is understood that I_C should be kept constant in spite of variations of I_{CBO} or I_{CO}. The extent to which a biasing circuit is successful in maintaining this is measured by stability factor. It is denoted by S.
* By definition, the rate of change of collector current I_C with respect to the collector leakage current I_{CO} at constant β and I_B is called stability factor. $S = 1$ is the ideal value.

$$S = \frac{dI_C}{dI_{CO}} \quad \text{at constant } I_B \text{ and } \beta$$

The general expression of stability factor for a CE configuration can be obtained as under.

$$I_C = \beta I_B + (\beta+1) I_{CO}$$

Differentiating above expression with respect to I_C, we get

$$1 = \beta \frac{dI_B}{dI_C} + (\beta+1) \frac{dI_{CO}}{dI_C}$$

$$1 = \beta \frac{dI_B}{dI_C} + \frac{(\beta+1)}{S}$$

since $\frac{dI_{CO}}{dI_C} = \frac{1}{S}$

OR

$$S = \frac{\beta + 1}{1 - \beta\left(\frac{dI_B}{dI_C}\right)}$$

Hence the stability factor S depends on β, I_B and I_C

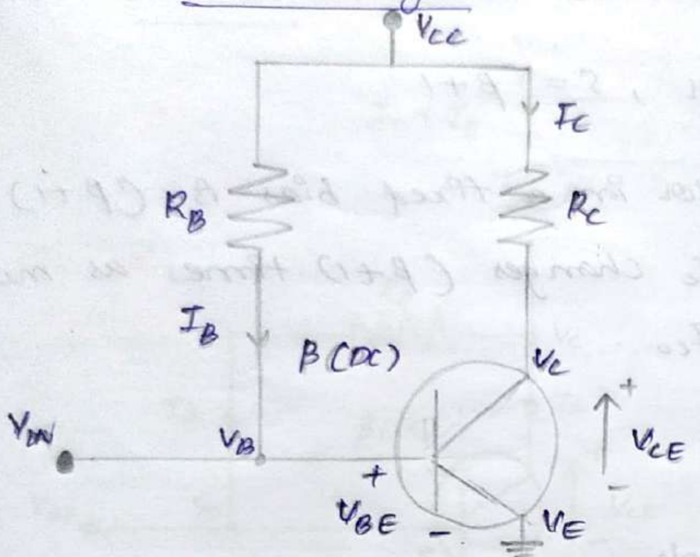
METHODS OF TRANSISTOR BIASING

The commonly used methods of transistor biasing are

- Base Resistor Method (Fixed bias)
- collector to Base bias
- Biasing with collector feedback resistor
- voltage - divider bias

All of these methods have the same basic principle of obtaining the required value of I_B and I_C from V_{CC} in the zero signal of conditions.

Fixed Base Biasing a Transistor



$V_C = V_{CC} - (I_C R_C)$

$V_{CE} = V_C - V_E$

$V_E = 0V$

$V_B = V_{BE}$

$I_B = \dfrac{V_{CC} - V_{BE}}{R_B}$

$I_C = \beta(DC) I_B$

$I_E = (I_C + I_B) \cong I_C$

The circuit shown is called as a "fixed base bias circuit", because the transistors base current, I_B remains constant for given values of V_{CC}, and therefore the transistors operating point must also remain fixed. This two resistor biasing network is used to establish the initial operating region of the transistor using a fixed current bias.

Stability factor

$$S = \frac{\beta+1}{1-\beta\left(\frac{dI_B}{dI_C}\right)}$$

In fixed-bias method of biasing, I_B is independent of I_C so that

$$\frac{dI_B}{dI_C} = 0$$

Substituting the above value in the previous equation,

Stability factor , $S = \beta + 1$

Thus the stability factor in a fixed bias is $(\beta+1)$ which means that I_c changes $(\beta+1)$ times as much as any change in I_{co}.

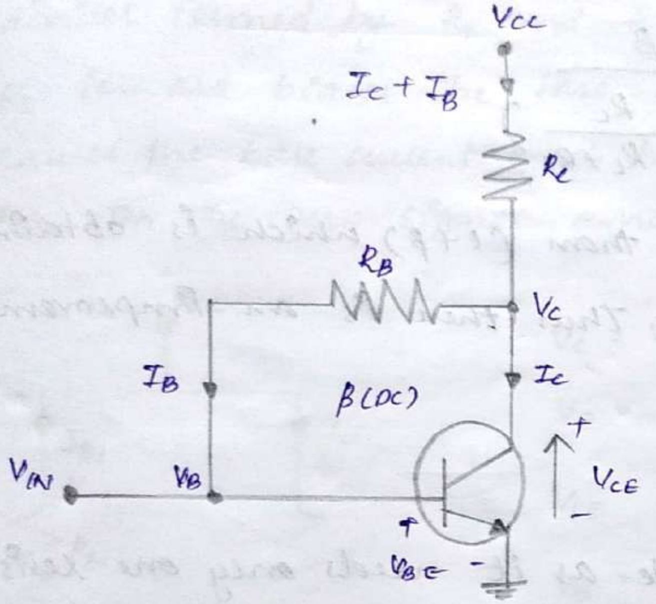
Advantages :

* The circuit is simple.
* Only one resistor R_E is required
* Biasing conditions are set easily.
* No loading effect as no resistor is present at base-emitter junction.

Disadvantages :

* The stabilization is poor as heat development can't be stopped.
* The stability factor is very high. So, there are strong chances of thermal runway. Hence this method is rarely employed.

Collector to Base Bias



$V_C = V_{CC} - R_C(I_C + I_B)$

$V_E = 0V$

$V_B = V_{BE}$

$I_B = \dfrac{V_C - V_B}{R_B}$

$I_C = \beta(DC)\, I_B$

$I_E = (I_C + I_B) \cong I_C$

* The collector to base bias circuit is same as base bias circuit except that the base resistor R_B is returned to collector, rather than to V_{CC} supply

* This circuit helps in improving the stability considerably. If the value of I_C increases, the voltage across R_L increases and hence the V_{CE} also increases. This inturn reduces the base current I_B. This action somewhat compensates the original increase.

$$\frac{dI_B}{dI_C} = -\frac{R_L}{R_L + R_B}$$

We know that

$$S = \frac{1+\beta}{1-\beta(dI_B/dI_C)}$$

Therefore

$$S = \frac{1+\beta}{1+\beta\left(\frac{R_C}{R_C+R_B}\right)}$$

This value is smaller than $(1+\beta)$ which is obtained for fixed bias circuit. Thus there is an improvement in the stability.

Advantages

* The circuit is simple as it needs only one resistor.
* The circuit provides some stabilization, for lesser changes.

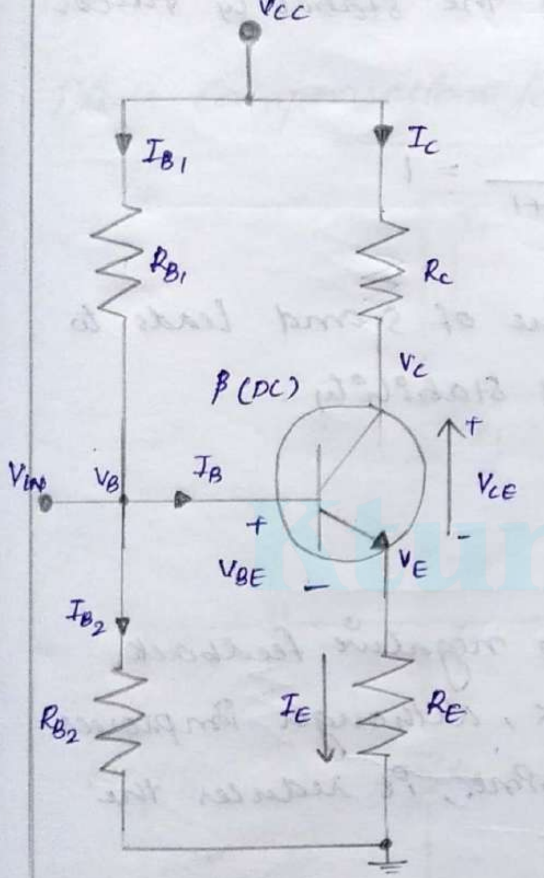
Disadvantages

* The circuit doesn't provide good stabilization.
* The circuit provides negative feedback.

Voltage Divider Bias Method

* Among all the methods of providing biasing and stabilization, the voltage divider bias method is the most prominent one. Here, two resistors R_1 and R_2 are employed, which are connected to V_{CC} and provide biasing. The resistor R_E employed in the emitter provides stabilization.

* The name voltage divider comes from the voltage divider formed by R_1 and R_2. The voltage drop across R_2 forward biases the base-emitter junction. This causes the base current and hence collector current flow in the zero signal conditions.



$$V_C = V_{CC} - I_C R_C = (V_E + V_{CE})$$

$$V_E = I_E R_E = V_B - V_{BE}$$

$$V_{CE} = V_C - V_E = V_{CC} - (I_C R_C + I_E R_E)$$

$$V_B = V_{BE} + V_E = V_{RB2} = \left(\frac{R_{B2}}{R_{B1} + R_{B2}}\right) V_{CC}$$

$$I_{B2} = \frac{V_B}{R_{B2}}$$

$$I_{B1} = I_B + I_{B2} = \frac{V_{CC} - V_B}{R_{B1}}$$

$$R_B = \frac{R_{B1} \times R_{B2}}{R_{B1} + R_{B2}}$$

$$I_C = \beta(DC)\, I_B \qquad I_B = \frac{V_B - V_{BE}}{R_B + (1+\beta) R_E}$$

$$I_E = I_C + I_B = \frac{V_E}{R_E}$$

The equation for stability factor of this circuit is obtained as

$$\text{Stability factor} = S = \frac{(\beta+1)(R_0 + R_3)}{R_0 + R_E + \beta R_E}$$

$$= (\beta+1) \times \frac{1 + R_0/R_E}{\beta + 1 + R_0/R_E}$$

where

$$R_0 = \frac{R_1 R_2}{R_1 + R_2}$$

If the ratio R_0/R_E is very small, then R_0/R_E can be neglected as compared to 1 and the stability factor becomes

$$\text{Stability factor} = S = (\beta+1) \times \frac{1}{\beta+1} = 1$$

This is the smallest possible value of S and leads to the maximum possible thermal stability.

BIAS COMPENSATION

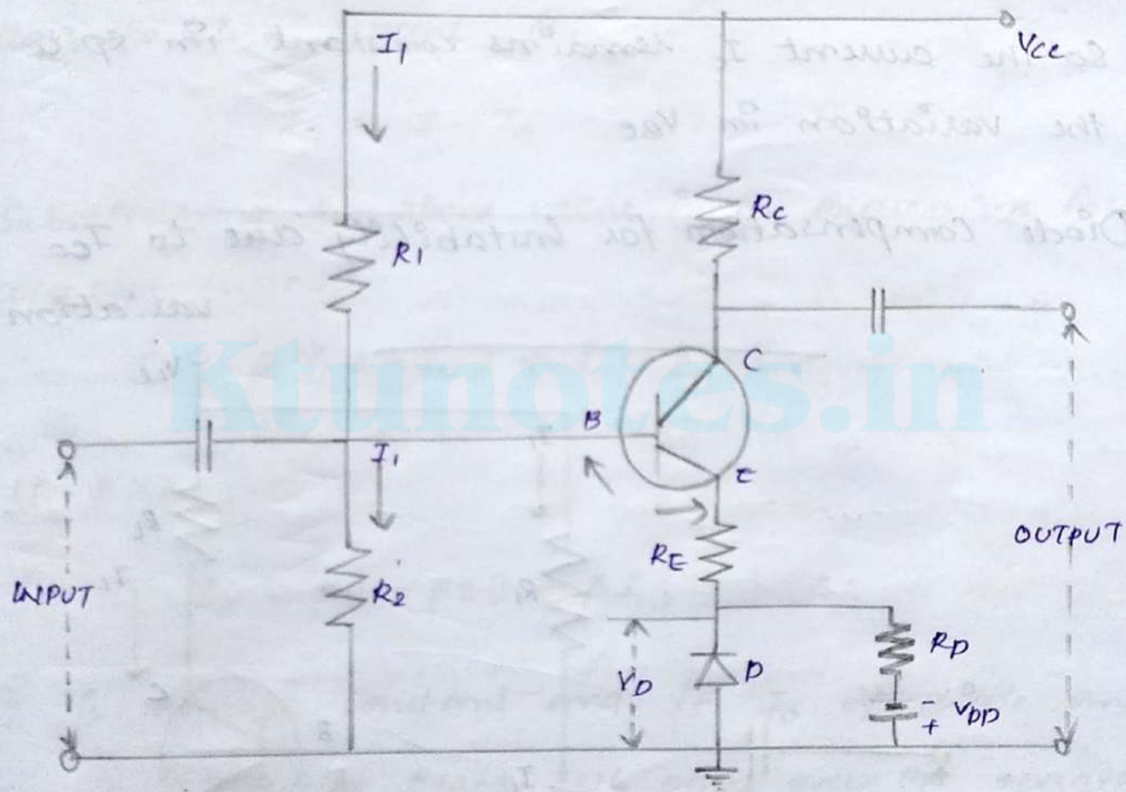
* The stabilization occurs due to negative feedback action. The negative feedback, although improves the stability of operating point, it reduces the gain of the amplifier.
* As the gain of the amplifier is a very important consideration, some compensation techniques are used to maintain excellent bias and thermal stabilization.
* Two types
  (i) Using diodes
  (ii) Using thermistors

Diode compensation for Instability

There are two types of diode compensation methods. They are

- Diode compensation for instability due to V_{BE} variation.
- Diode compensation for instability due to I_{co} variation.

Diode Compensation for Instability due to V_{BE} variation



* In a silicon transistor, the changes in the value of V_{BE} results in the changes in I_c. A diode can be employed in the emitter circuit in order to compensate the variations in V_{BE} or I_{CO}. As the diode and transistor used are of same material,

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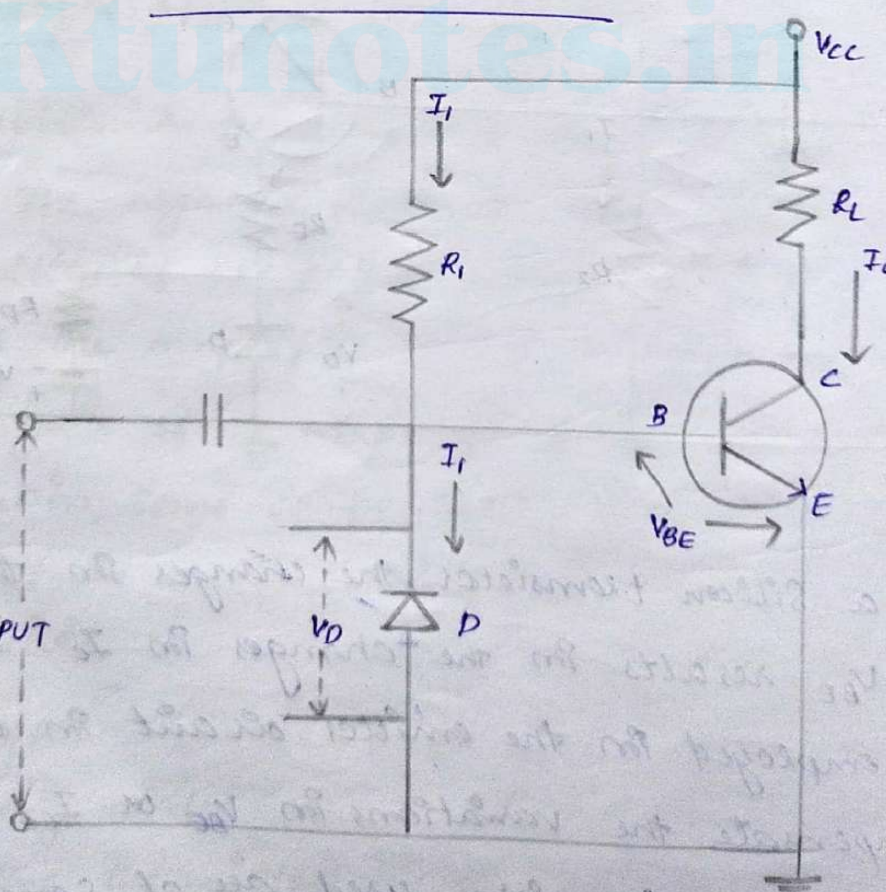
The voltage V_D across ... temperature coefficient as V_{BE} of the transistor

* The diode D is forward biased by the source V_{DD} and the resistor R_D. The variation in V_{BE} with temperature is same as the variation in V_D with temperature, hence the quantity $(V_{BE} - V_D)$ remains constant. So the current I_C remains constant. So the current I_C remains constant in spite of the variation in V_{BE}.

Diode Compensation for Instability due to I_{CO} variation



So, the reverse saturation current I_0 of the diode will increase with temperature at the same rate as the transistor collector saturation current I_{CO}.

$$I = \frac{V_{CC} - V_{BE}}{R} \cong \frac{V_{CC}}{R} = \text{constant}$$

The diode D is reverse biased by V_{BE} and the current through it is the reverse saturation current I_0.

Now the base current is,

$$I_B = I - I_0$$

Substituting the above value in the expression for collector current

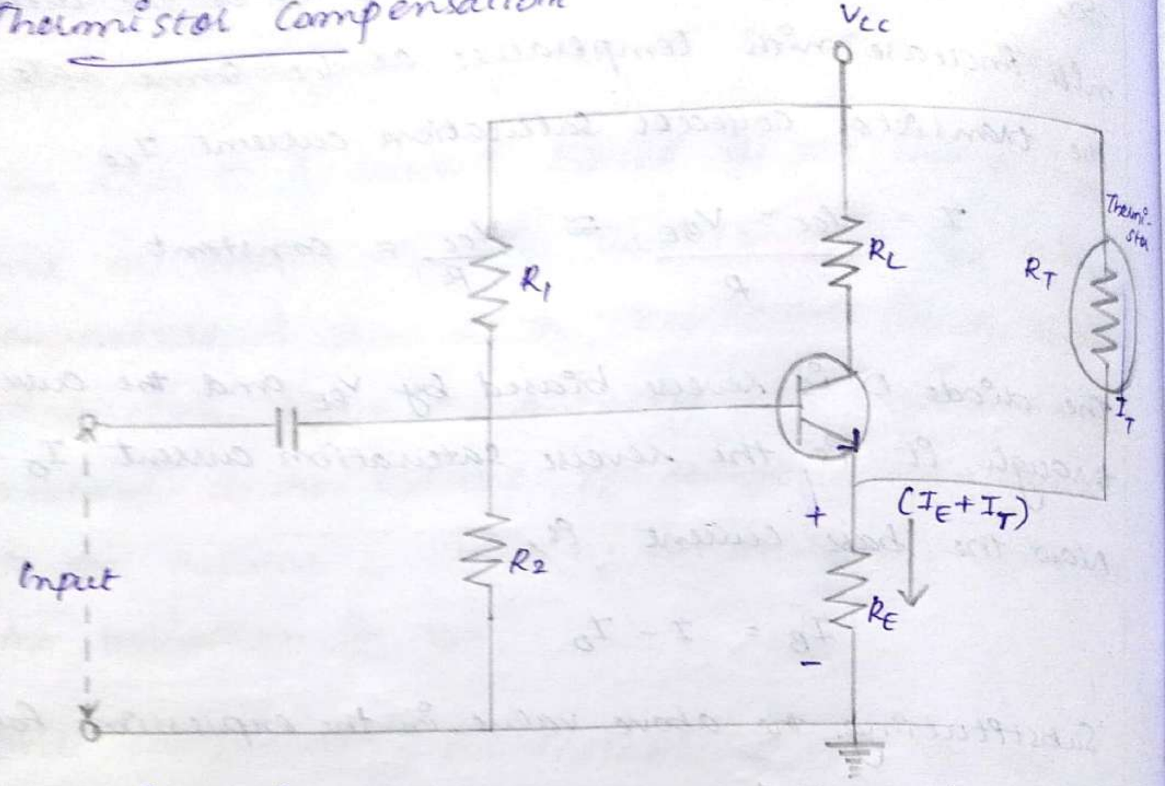
$$I_C = \beta(I - I_0) + (1+\beta) I_{CO}$$

If $\beta >> 1$,

$$I_C = \beta I - \beta I_0 + \beta I_{CO}$$

I is almost constant and if I_0 of diode and I_{CO} of transistor track each other over the operating temperature range, then I_C remains constant.

Thermistor Compensation



* Thermistor is a temperature sensitive device. It has negative temperature coefficient. The resistance of a thermistor increases when the temperature decreases and it decreases when the temperature increases.

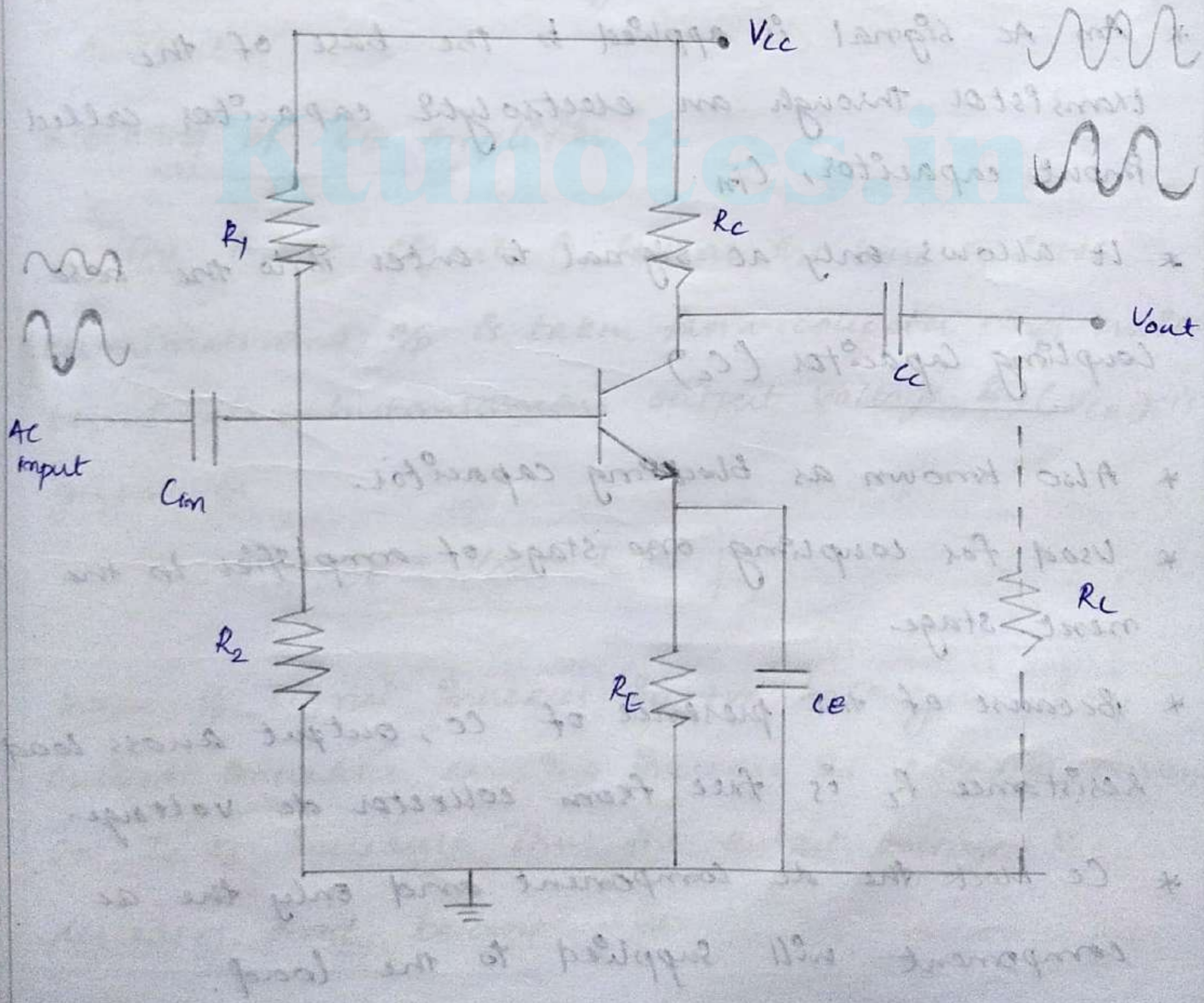
* In an amplifier circuit, the changes that occur in I_{co}, V_{BE} and β with temperature, increases the collector current. Thermistor is employed to minimize the increase in collector current. As the temperature increases, the resistance R of thermistor decreases, which increases the current through it and the resistor R. Now, the voltage developed across R increases, which reverse biases the emitter junction. This reverse

bias is so high that the effect of resistors R_1 and R_2 providing forward bias also gets reduced. This action reduces the rise in collector current.

* Thus the temperature sensitivity of thermistor compensates the increase in collector current, occurred due to temperature.

Common Emitter Amplifier

Consider a practical circuit of single stage transistor amplifier in common emitter (CE) configuration.



→ $R_1 - R_2$ voltage divider bias circuit

* R_1 and R_2 are employed to provide the necessary biasing. These resistors setup the Q point in the middle of the load line.

* R_E provides perfect stabilization. ie, it doesnot allow the Q point to shift its position due to rise in temperature or due to change in transistor parameters.

→ Input Capacitor, C_{in}

* An AC signal is applied to the base of the transistor through an electrolyte capacitor called input capacitor, C_{in}.

* It allows only ac signal to enter into the base.

→ Coupling Capacitor (C_c)

* Also known as blocking capacitor.
* Used for coupling one stage of amplifier to the next stage.
* Because of the presence of Cc, output across load resistance R_L is free from collector dc voltage.
* Cc block the dc component and only the ac component will supplied to the load.

* In the absence of this capacitor, R_C will become parallel with R_1 and there by change the biasing condition of next stage.

Emitter Bypass Capacitor (C_E)

* When ac signal is applied to an amplifier circuit, variable current will flow through R_C and R_E. This current in R_E will develop a negative feedback to the emitter junction. This will result in the reduction of overall voltage gain of the amplifier.

* Capacitor C_E prevents negative feedback in the emitter current.

Working of CE amplifier

The input signal is fed at the base emitter terminals and o/p is taken from collector and emitter terminals. Instantaneous output voltage (V_{CE}) is given as

$$\boxed{V_{CE} = V_{CC} - I_C R_C}$$

When i/p signal increases in +ve half cycle, base current increases, causing increase in collector current. So $I_C R_C$ increases. Thus the output voltage V_{CE} decreases and become -ve.

When i/p signal increases in the half cycle, Base current ↑, I_C ↑, $I_C R_C$ ↓, Then o/p V_{CE} ↑

There is 180° phase shift between i/p and o/p wave

Small signal CE amplifier circuit using h parameters

A transistor can be treated as a two port network. The behaviour of any two point n/w can be specified by

→ Terminal voltages V_1 and V_2 at ports 1 and 2

→ Currents entering port i_1 and i_2 at ports 1 and 2.

* Of the four variables, V_1, V_2 and i_2 two variables can be selected as independent variables and other two are dependent variables which can be expressed in terms of independent variables.

* The different two port parameters are:

(i) Z parameters or impedance parameters.

(ii) Y parameters or admittance parameters

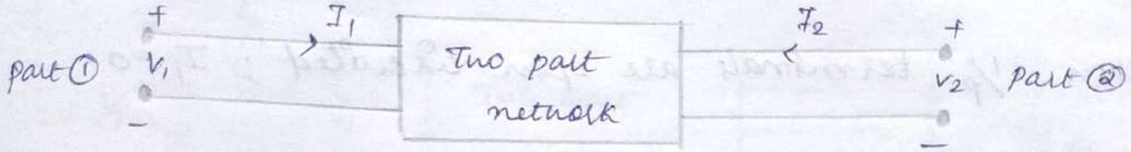
(iii) h parameters or hybrid parameters.

* For transistors hybrid parameters are used.

* The four parameters which are used to analyse any linear circuit having input and output

terminals are called hybrid parameters or h-parameters.

Consider a general two port network



V_1 and I_2 are dependent variables.

V_2 and I_1 are independent variables.

$\therefore V_1 = f(I_1, V_2)$

$I_2 = f(I_1, V_2)$

$$\boxed{\begin{aligned} V_1 &= h_{11} I_1 + h_{12} V_2 \quad \text{——— ①} \\ I_2 &= h_{21} I_1 + h_{22} V_2 \quad \text{——— ②} \end{aligned}}$$

Calculation of h parameters

→ When o/p terminals are short circuited; $V_2 = 0$

$\therefore h_i = h_{11} = \dfrac{V_1}{I_1}$; unit is ohm
i/p resistance when o/p port is short circuited.

$h_f = h_{21} = \dfrac{I_2}{I_1}$; dimension less
forward current gain when o/p port short circuited.

For CE configuration

$$h_i = h_{ie}$$

$$h_f = h_{fe}$$

when i/p terminals are open circuited ; $I_1 = 0$

$h_r = h_{12} = \frac{V_1}{V_2}$; dimension less

Reverse voltage gain when i/p part is opened.

$h_o = h_{22} = \frac{I_2}{V_2}$; unit is mho

o/p admittance when i/p part opened.

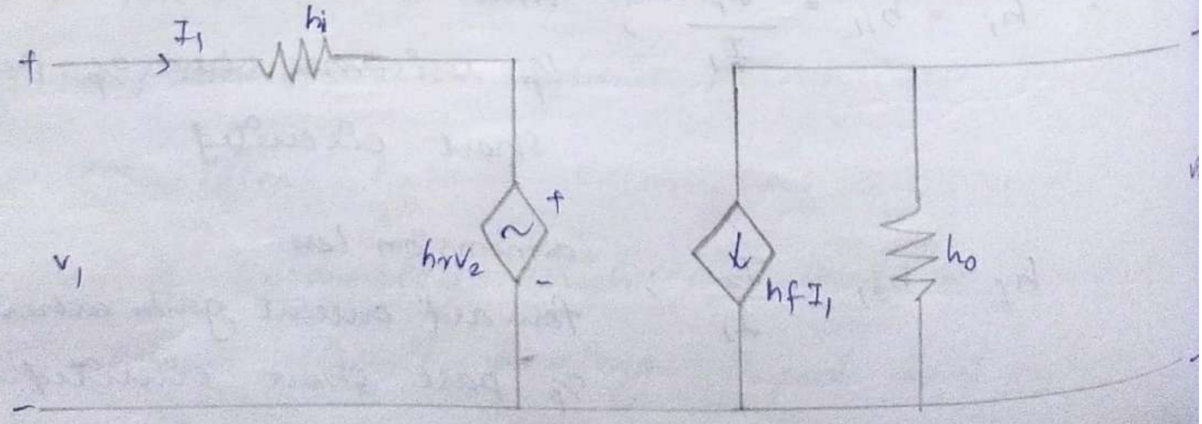
∴ The four h parameters are h_i, h_o, h_r, h_f

we replace equations ① and ② by above four parameters and we get

$$V_1 = h_i I_1 + h_r V_2 \quad \text{——} ③$$

$$I_2 = h_f I_1 + h_o V_2 \quad \text{——} ④$$

∴ The equivalent ckt can be obtained as



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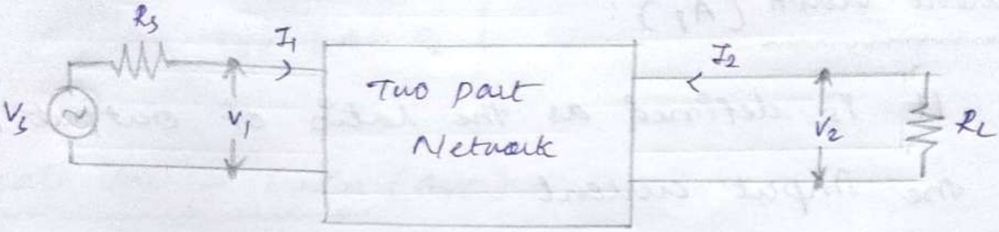
→ To form a transistor amplifier it is necessary to connect an external load and signal source as shown in the figure.



For CE amplifier

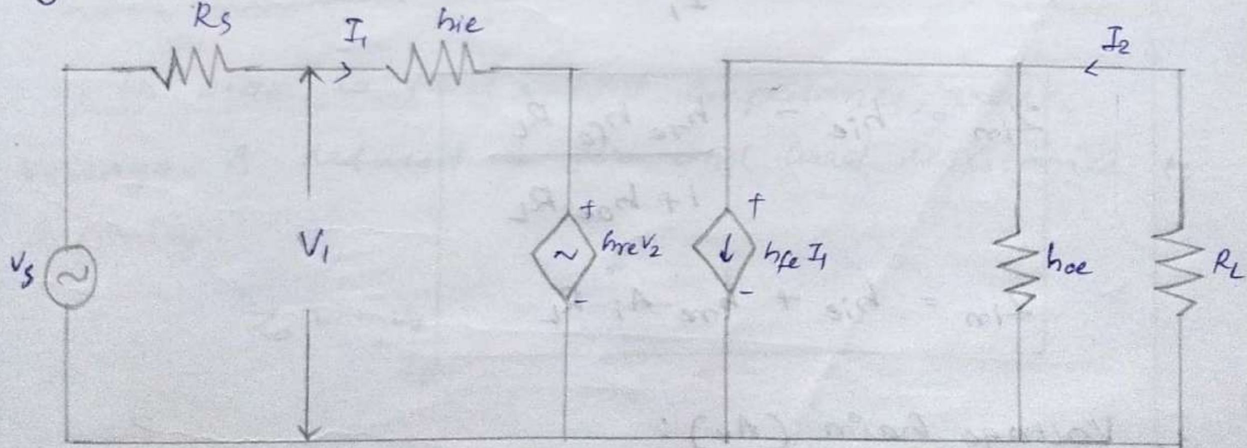
$$h_i = h_{ie}$$

$$h_r = h_{re}$$

$$h_o = h_{oe}$$

$$h_f = h_{fe}$$

Hybrid model for CE configuration is represented as



* h parameters equations of the circuit are

$$\boxed{\begin{aligned} V_1 &= h_{ie} I_1 + h_{re} V_2 \\ I_2 &= h_{fe} I_1 + h_{oe} V_2 \end{aligned}}$$

Amplifier Gains and Impedance calculations using h- equivalent circuit

1. Current Gain (A_i):

It is defined as the ratio of output current to the input current.

$$A_i = \frac{-I_2}{I_1}$$

$$\boxed{A_i = \frac{-h_{fe}}{1+h_{oe}R_L}}$$

2. Input Impedance (Z_{in}):

$$Z_{in} = \frac{V_1}{I_1}$$

$$\boxed{\begin{aligned} Z_{in} &= h_{ie} - \frac{h_{re}h_{fe}R_L}{1+h_{oe}R_L} \\ Z_{in} &= h_{ie} + h_{re}A_iR_L \end{aligned}}$$

3. Voltage Gain (A_v):

Ratio of output voltage (V_2) to input voltage (V_1) is known as voltage gain.

$$A_V = \frac{V_2}{V_1}$$

$$\boxed{A_V = \frac{-h_{fe} R_L}{h_{ie} + \Delta h R_L}}$$ $\Delta h = h_{ie} h_{oe} - h_{re} h_{fe}$

4. Overall Voltage Gain (A_{VS}):

Ratio of output voltage V_2 to source voltage V_s is known as overall voltage gain.

$$A_{VS} = \frac{V_2}{V_S}$$

$$\boxed{A_{VS} = \frac{A_i R_L}{R_S + Z_{in}}}$$

5. Output Impedance (Z_o):

In order to find output impedance, source voltage is reduced to zero and load resistance to infinity.

$$Z_o = \frac{V_2}{I_2}$$

$$\boxed{Z_o = \frac{R_s + h_{ie}}{\Delta h + h_{oe} R_s}}$$ $\Delta h = h_{ie} h_{oe} - h_{re} h_{fe}$

If R_s (source resistance) is zero, $R_s = 0$

$$\therefore \quad Z_o = \frac{h_{ie}}{\Delta h}$$

6. Overall Current Gain :

Ratio of output current (I_L) to the current delivered by the source (I_s) is known as overall current gain.

$$A_{is} = \frac{I_L}{I_s} = \frac{-I_2}{I_s}$$

$$\boxed{A_{is} = \frac{A_i R_s}{R_s + Z_{in}}}$$

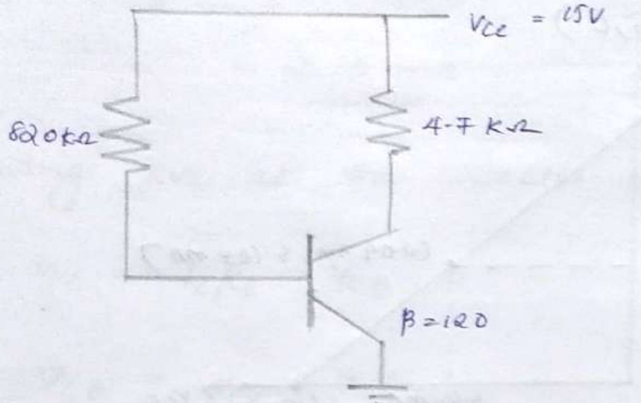
7. Power Gain :

Power gain is the product of voltage gain and current gain.

$$\boxed{A_p = A_v \cdot A_i}$$

PROBLEMS

1. Draw the dc load line and locate the operating point for fixed biasing circuit shown in figure. What will be its stability factor.



Ans:- For a transistor NPN $V_{BE} = 0.7 V$. To find load line we have to calculate I_C and V_{CE}. From I_B we get I_C.

Applying kirchoff's voltage law to the i/p side.

$$V_{CC} = I_B R_B + V_{BE}$$

$$I_B = \frac{V_{CC} - V_{BE}}{R_B} = \frac{15 - 0.7}{820 \times 10^3 \Omega} = \underline{17.44 \mu A}$$

$$I_C = \beta I_B = 120 \times 17.44 \mu A = \underline{2.09 mA}$$

Collector Emitter voltage, $V_{CE} = V_{CC} - I_C R_C$

$$= 15 - (2.09 mA) \times 4.7 k\Omega$$

$$= \underline{5.164 V}$$

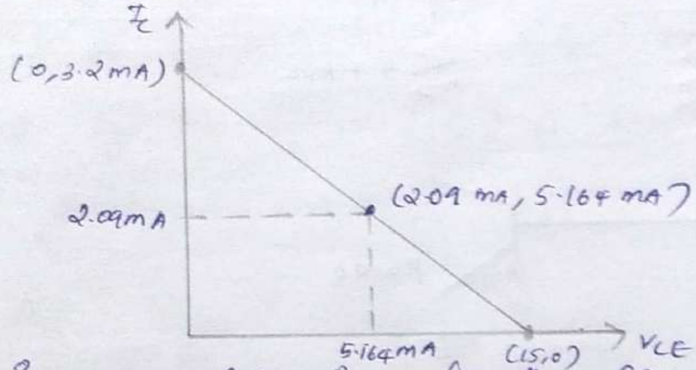
Stability factor $S = 1 + \beta = \underline{121}$

For dc load line, point A is $(0, \frac{V_{CC}}{R_C})$

and point B $(V_{CC}, 0)$

points are A $(0, \frac{15}{4.7K\Omega}) = (0, 3.2 mA)$

point B $(15, 0)$

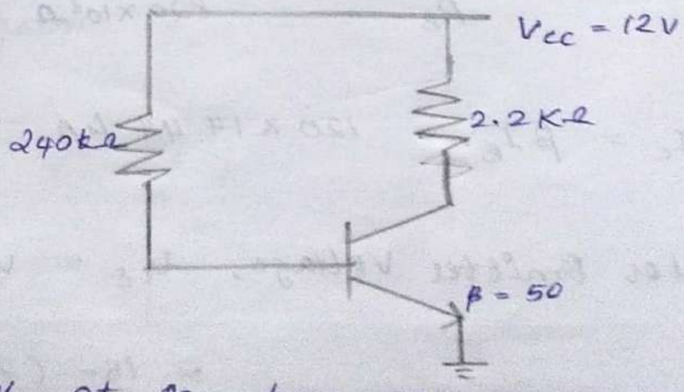


2. Determine the following for the fixed bias configuration for the transistor

ⓐ I_B and I_C ⓑ V_{CE} ⓒ V_B and V_C ⓓ V_{BC}

Given $\beta = 50$, $V_{CC} = 12V$, $R_C = 2.2K\Omega$, $R_B = 240K\Omega$

Ans:- First of all draw the fixed bias circuit with given values



(a) Applying KVL at the base ckt

$$V_{CC} = I_B R_B + V_{BE}$$

$$I_B = \frac{V_{CC} - V_{BE}}{R_B}$$ $V_{BE} = 0.7V$ for si

$$= \frac{12 - 0.7}{240 \times 10^3} = \underline{\underline{48.08\ \mu A}}$$

Then $I_C = \beta I_B = 50 \times 48.08$

$$= \underline{\underline{2.35\ mA}}$$

(b) Applying KVL at the collector ckt

$$V_{CC} = I_C R_C + V_{CE}$$

$$V_{CE} = V_{CC} - I_C R_C$$

$$= 12 - [2.35 \times 10^{-3} \times 2.2 \times 10^{+3}]$$

$$= \underline{\underline{6.83V}}$$

(c) $V_B - V_{BE} = 0$ Similarly $V_C - V_{CE} = 0$

$V_B = V_{BE} = 0.7V$ $V_C = V_{CE} = \underline{\underline{6.83V}}$

V_B is the voltage b/w base and ground

V_C is the voltage b/w collector and ground

(d) $V_{BC} = V_B - V_C$

$$= 0.7 - 6.83$$

$$= \underline{\underline{-6.13V}}$$

3. Design a voltage divider bias circuit to operate from a 12 V supply. The given bias conditions are $V_{CE} = 3V$, $V_E = 5V$ and $I_C = 1mA$

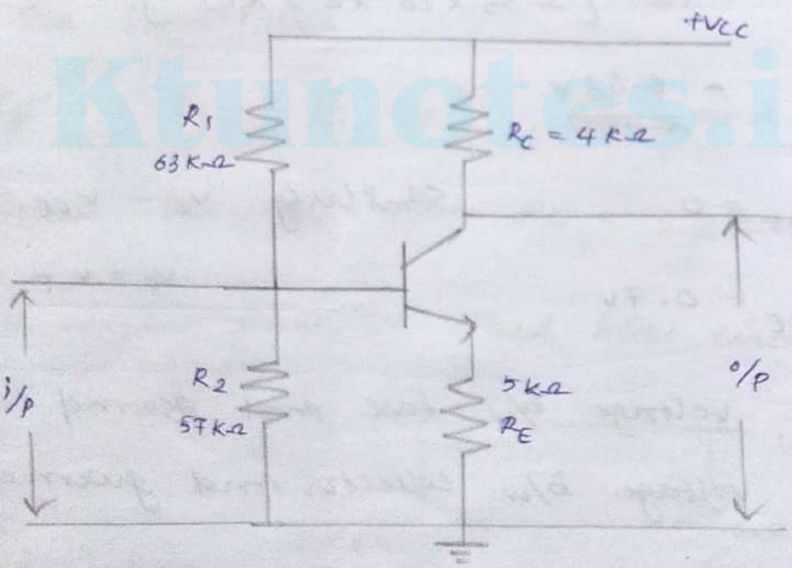
Ans:- Assumptions required for designing resistors for voltage divider bias circuit.

→ Select a NPN silicon transistor. So that $V_{BE} = 0.7V$

→ $I_C \simeq I_E$ ∴ I_B is very small

→ $I_{R_1} = I_{R_2}$

{ β is not given for voltage bias ckt



(i) Design of R_E

$$V_{RE} = V_E = 5V \text{ (Given)}.$$

ie $I_E R_E = 5V$

$$\therefore R_E = \frac{5V}{I_E} = \frac{5V}{1mA} = 5K\Omega$$

Since $I_E \simeq I_C$

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(ii) Design of R_C

Apply KVL to the collector ckt

i.e, $V_{CC} = I_C R_C + V_{CE} + V_E$

$$I_C R_C = V_{CC} - V_{CE} - V_E$$

$$R_C = \frac{V_{CC} - V_{CE} - V_E}{I_C} = \frac{12-3-5}{1mA} = \frac{4}{1mA} = \underline{4k\Omega}$$

(iii) Design of R_1 and R_2

$$V_{R_2} = V_B = V_{BE} + V_E = 0.7 + 5 = \underline{5.7V}$$

$$I_C = 10 I_{R_1} = 10 I_{R_2}$$

$$I_{R_2} = \frac{I_C}{10} = \underline{0.1 mA}$$

$$V_{R_2} = I_{R_2} \cdot R_2$$

$$R_2 = \frac{V_{R_2}}{I_{R_2}} = \frac{5.7}{0.1mA} = \underline{57 k\Omega}$$

$$V_{R_1} = V_{CC} - V_B = V_{CC} - V_{R_2} = 12 - 5.7V = \underline{6.3V}$$

$$I R_1 = V_{R_1}$$

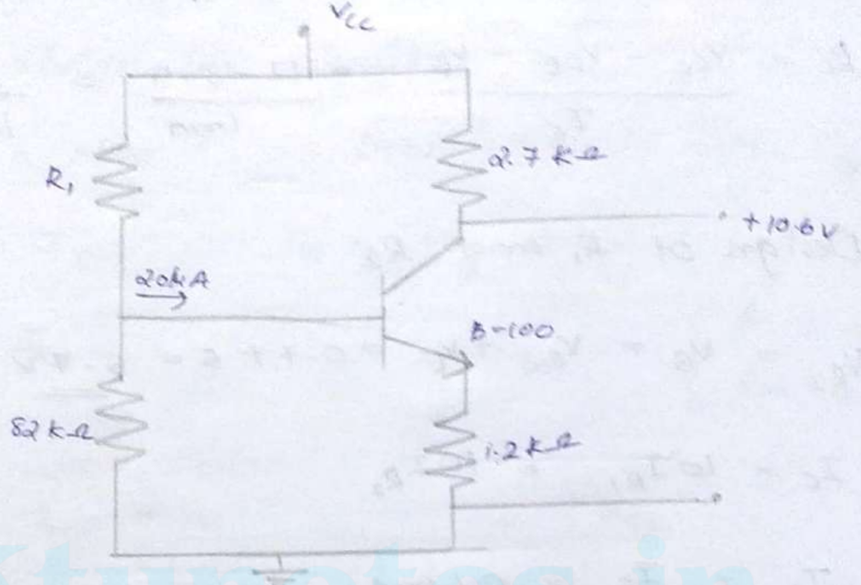
$$\therefore R_1 = \frac{V_{R_1}}{I_{R_1}} = \frac{6.3}{0.1mA} = \underline{63 k\Omega}$$

Substitute these values in the voltage divider bias ckt.

4. For a voltage divider bias configuration of the fig shown below, determine

(i) I_C (ii) V_E (iii) V_{CC} (iv) V_{CE}

(v) V_B (vi) R_1



Ans:- From the figure $I_B = 20\mu A$ $R_2 = 82\,k\Omega$

$V_C = 10.6\,V$ $R_C = 2.7\,k\Omega$

$\beta = 100$ $R_E = 1.2\,k\Omega$

(i) For finding I_C :-

Base current $I_B = 20\mu A$

Collector current $I_C = \beta I_B = 100 \times 20\mu A$

$= \underline{2\,mA}$

(ii) To find V_E :-

Emitter current $I_E = I_B + I_C$

$= 2mA + 20\mu A$

$= \underline{\underline{2.02\ mA}}$

Emitter voltage $V_E = I_E \cdot R_E$

$= 2.02\ mA \times 1.2 K\Omega$

$= \underline{\underline{2.424\ V}}$

(iii) To find V_{CC} :-

Supply voltage $V_{CC} = V_C + I_C R_C$

$= 10.6\ V + (2mA \times 82 K\Omega)$

$= \underline{\underline{16\ V}}$

(iv) To find V_{CE} :-

$V_{CE} = V_C - V_E = 10.6 - 2.424 = \underline{\underline{8.176\ V}}$

(v) Base voltage $V_B = V_E + V_{BE} = 2.424 + 0.7V$

$= \underline{\underline{3.124\ V}}$

(vi) To find R_1

By applying Thevenin's voltage, $V_{TH} = \frac{R_2}{R_1 + R_2} \cdot V_{CC} = V_B$

$$V_{TH} = V_B = \frac{8.2 K\Omega}{R_1 + 8.2 K\Omega} \times 16 = \underline{\underline{3.124\ V}}$$

$$\therefore R_1 = \frac{8.2 \times 16}{3.124} - 8.12 = \underline{\underline{33.8\,K\Omega}}$$